

Section 4.9 Anti-Derivatives

Given $y = x^2 + 5$ find y' .

$$y' = 2x$$

Anti-Derivative means that we work backwards: given y' , find y .

Ex ① Given $y' = x^7 + 4$, find y .

$$y = \frac{x^{7+1}}{8} + \frac{4x^{0+1}}{1} + C \quad \text{constant}$$

$$y = \frac{x^8}{8} + 4x + C$$

Take the derivative to check:

$$y' = \frac{8x^7}{8} + 4 + 0$$

$$y' = x^7 + 4 \quad \checkmark$$

$$\text{If } y' = x^n$$

$$\text{then } y = \frac{x^{n+1}}{n+1} + c$$

Also, use the derivatives you have memorized to help find anti-derivatives.

$$\text{Ex } \textcircled{2} \quad y' = e^x + \frac{1}{2\sqrt{x}} - \frac{1}{x}$$

$$y = e^x + \sqrt{x} - \ln x + c$$

Sometimes it helps to do Algebra first.

$$\text{Ex } \textcircled{3} \quad y' = \frac{x^2 - \sqrt{x} + 1}{\sqrt{x}} \quad \text{separate it into 3 fractions.}$$

$$y' = \frac{x^2}{\sqrt{x}} - \frac{\sqrt{x}}{\sqrt{x}} + \frac{1}{\sqrt{x}} \quad \text{write each one as } x^n$$

$$y' = x^{\frac{3}{2}} - x^0 + x^{-\frac{1}{2}}$$

$$y = \frac{x^{\frac{3}{2}+1}}{\frac{5}{2}} - \frac{x^{0+1}}{1} + \frac{x^{-\frac{1}{2}+1}}{\frac{1}{2}} + C$$

$$y = \frac{2}{5} x^{\frac{5}{2}} - x + 2x^{\frac{1}{2}} + C$$

Ex ④ $y' = (3x-1)(2x+5)$

Since we do not have a product rule for anti-derivatives (yet), we need to multiply this out first.

$$y' = 6x^2 + 15x - 2x - 5$$

$$y' = 6x^2 + 13x - 5$$

$$y = \frac{6x^3}{3} + \frac{13x^2}{2} - 5x + C$$

$$y = 2x^3 + \frac{13}{2}x^2 - 5x + C$$

Sometimes you will be given number to plug in so that you can solve for c .

Ex ⑤ $y' = \sin x + \cos x$ and $y(0) = 3$
↳ "initial conditions"

$$y = -\cos x + \sin x + c$$

Now substitute $x=0$ and $y=3$.

$$3 = -\cos(0) + \sin(0) + c$$

$$3 = -1 + 0 + c$$

$$4 = c$$

$$y = -\cos x + \sin x + 4$$

We can do this with second derivatives also!

Ex ⑥ $y'' = x^2 + 3$ and $y'(1) = 2$, $y(0) = 7$.

$$y' = \frac{x^3}{3} + 3x + c$$

Substitute $x=1$ and $y'=2$ to find c .

$$2 = \frac{1^3}{3} + 3 + c$$

$$-\frac{4}{3} = c$$

$$y' = \frac{x^3}{3} + 3x - \frac{4}{3}$$

$$y = \frac{x^4}{12} + \frac{3x^2}{2} - \frac{4}{3}x + k$$

Substitute $x=0$ and $y=7$ to find k .

$$7 = \frac{0^4}{12} + \frac{3(0)^2}{2} - \frac{4}{3}(0) + k$$

$$7 = k$$

$$y = \frac{x^4}{12} + \frac{3x^2}{2} - \frac{4}{3}x + 7$$

Take the second derivative and make sure that $y'' = x^2 + 3$

$$y' = \frac{4x^3}{12} + \frac{6x}{2} - \frac{4}{3}$$

$$y' = \frac{x^3}{3} + 3x - \frac{4}{3}$$

$$y'' = \frac{3x^2}{3} + 3 - 0$$

$$y'' = x^2 + 3 \quad \checkmark$$

What About the Chain Rule?

$$\text{If } y = e^{5x}, \text{ then } y' = e^{5x} \cdot 5$$

↑
Chain Rule
for derivatives

$$\text{If } y' = e^{5x}, \text{ then } y = \frac{e^{5x}}{5} + c$$

↑
divide because
we are doing the
opposite of the
derivative.

Ex ⑦ $y' = \frac{1}{1+4x^2}$

Recall $\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}$

So the answer is probably $\tan^{-1}$
but what about the 4?

$y' = \frac{1}{1+4x^2}$ can be rewritten as

$y' = \frac{1}{1+(2x)^2}$ it must be $\tan^{-1}(2x)$

$y = \frac{\tan^{-1}(2x)}{2} + C$
2 ← because of the Chain Rule

Take the derivative and see

if $y' = \frac{1}{1+4x^2}$

$$y = \frac{\tan^{-1}(2x)}{2} + C$$

$$y' = \frac{1}{2} \left(\frac{1}{1 + (2x)^2} \right) \cdot 2 + 0$$

chain Rule

$$y = \frac{1}{1 + 4x^2} \quad \checkmark$$